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- **Problem Statement ID –** 26167
- **Problem Statement Title –** SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme –** Space Technology
- **PS Category –** Software
- **Team ID –** ____________ (from portal)
- **Team Name (Registered on portal) –** x64

Ask your satellite in plain language: a plan over specialist tools, a grounded answer, and a trace any reviewer can audit.

5 / 5 PS demo queries covered verbatim	**15** PS clauses mapped (P1–P15)	**3** models fine-tuned by 20 Sep	**Rs 0** licence · offline · laptop GPU

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IDEA TITLE

Proposed solution (idea / solution / prototype) · detailed explanation · how it addresses the problem · innovation and uniqueness

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PROPOSED SOLUTION / APPROACH

- **Agentic by design —** a natural-language query is planned into a sequence of specialist tools chosen from a predefined registry (P6–P8); never one monolithic VLM.
- **Guardrailed execution —** task schema plus parameter whitelist, then a deterministic router. Incompatible rasters return a compatibility report instead of a crash.
- **Auditable trace —** each run logs task, tool + version, permitted parameters, input hash, per-step latency and confidence (P9–P10) — the surface the PS says it grades.
- **Evidence on the image —** change mask, grounding boxes, caption overlay; calibrated High/Med/Low badges, an optical/SAR disagreement flag, and reasoned abstention.
- **RS-adapted, not generic —** Qwen2-VL-2B adapted with QLoRA on BigEarthNet.txt. A generic VLM “will not satisfy the requirements” (P12), so the graded path runs our weights, offline.

How it addresses the problem: *the PS asks for a system that plans, not a model that guesses. We turn an expert GIS pipeline into one query box and make every answer checkable line by line — closing the trust gap that keeps satellite imagery locked inside specialist desks.*

The five PS queries we demonstrate, verbatim — and what each routes to

1. *“Describe the land-cover and major objects visible in this image.”* » **caption + VQA**
2. *“Highlight the water body referred to in the query.”* » **text-guided grounding**
3. *“What changed between these two dates, and where did the change occur?”* » **bi-temporal change + change-VQA**
4. *“Use the optical and SAR images together to identify built-up and water-covered regions.”* » **cross-modal joint reasoning**
5. *“Has the built-up area increased, decreased, or remained unchanged?”* » **change-VQA + area trend**

END-TO-END WORKFLOW · every step lands in trace.json

Step		
1	**Query + imagery**	one optical/SAR raster or a co-registered bi-temporal pair; PNG/JPEG only for the named benchmarks (P11).
2	**Ingest & validate**	band order · CRS · modality · co-registration offset · 512 px tiles / 64 overlap, so resolution never changes code (P14).
3	**Plan & guard**	task id and tool set from registry.yaml; only whitelisted parameters are permitted; RuleRouter guarantees a valid plan.
4	**Execute specialists**	M1 VLM (single-image + 2-image) · M4 change · M3 grounding · M7 land-cover; every call writes to the trace.
5	**Fuse · answer · report**	geometric-mean confidence, badges, overlay raster, answer card, downloadable PDF/HTML audit pack.

Chain-of-thought is deliberately not shown: the PS evaluates only the observable execution trace, so a plan is audited by what it ran with, not by prose (P10). A deterministic RuleRouter therefore scores identically to an LLM planner — and never fails on stage.

INNOVATION AND UNIQUENESS · the five things another team cannot bolt on at the end

Trace as a product
the graded surface is a first-class UI, JSON and PDF artifact, not a debug log

Registry + guardrails
declared tools, whitelisted parameters, deterministic fallback: the demo cannot crash

Optical × SAR jointly
spectra plus all-weather structure; paired reasoning is the PS's principal focus (P1, P5)

RS-adapted weights
QLoRA on BigEarthNet.txt over S1/S2/fused backbones — never a hosted generic API

Honest confidence
calibrated badges, disagreement flag, abstention when the evidence is thin

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TECHNICAL APPROACH

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Technologies to be used (languages, frameworks, hardware) · methodology and process for implementation (flow charts / images / working prototype)

ZONE 1 · INPUTS & VALIDATION

Analyst / officer
one plain-language question; no QGIS, no scripting

Co-registered pair
Cartosat-2S optical + RISAT SAR (Sentinel proxies in dev)

Ingest gate (GDAL)
bands, CRS, modality, co-registration offset; PNG/JPEG only for named benchmarks

512 px tiles, 64 overlap
resolution- and sensor-agnostic by construction

→

ZONE 2 · WEB FRONTEND

Query console
upload, demo scenes, query chips, cached-answer switch

Answer card
answer, badges, confidence, abstention copy

Overlay stage
change mask / boxes / caption, toggled per layer

Trace panel
expandable steps + per-step latency + trace.json / PDF

→

ZONE 3 · AGENTIC BACKEND

API gateway (FastAPI)
auth, job queue, one endpoint per tool contract

Planner + guardrail
registry.yaml selection, schema and parameter whitelist

RuleRouter fallback
deterministic routing when the planner is unsure

Trace store (SQLite)
inputs hash, tools, versions, params, latency, confidence

→

ZONE 4 · INTELLIGENCE LAYER

M1 Qwen2-VL-2B QLoRA
VQA, captioning, 2-image optical+SAR — the backbone

M4 BiT change · M3 Grounding-DINO-T
bi-temporal map, change-VQA, text-to-box

M7 land-cover classifier
BigEarthNet v2.0 encoder; built-up, water, crop

Adapted on BigEarthNet.txt
9.55 M image-text rows incl. S1-S2-text triplets (P2)

TECHNOLOGY STACK | Python 3.11 | FastAPI | Streamlit / React | GDAL · rasterio | PyTorch + QLoRA | Qwen2-VL-2B | Grounding DINO-T | BiT · UNet++ | SQLite trace | Docker (offline)

SAR IS NOT RGB · what generic VLM pipelines skip

- **Radiometry:** backscatter stays in dB, with the stretch logged.
- **Speckle:** log transform + refined Lee filter before any patch is cut.
- **Polarisation:** VV / VH stay separate channels, never averaged.
- **Encoders:** optical branch from BE v2.0 S2 weights, SAR branch from S1, fused from S1+S2 — that trio is also the ablation.

MODELS × DATA · why this is not a GPT wrapper

M1 **Qwen2-VL-2B + QLoRA 4-bit** · BigEarthNet.txt · VRSBench · CDVQA · RSVQA

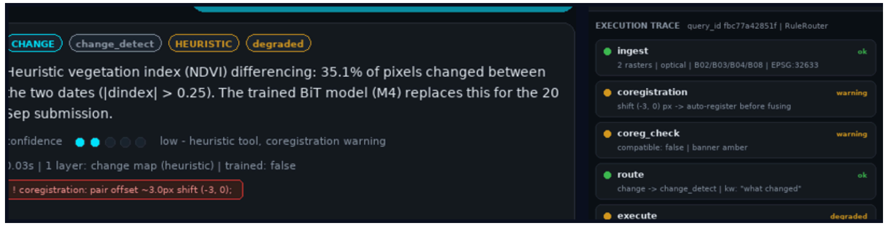
M3 **Grounding DINO-T** · VRSBench boxes · BE v2.0 reference maps

M4 **BiT encoder + change head** · LEVIR-CD · WHU-CD · CDVQA pairs

M7 **resnet50, BE v2.0 pretrained** · S1 / S2 / S1+S2, 19 land-cover classes

Training mix is weighted to paired samples: the PS names joint cross-modal and bi-temporal reasoning as the principal focus (P1).

RUNNING PROTOTYPE · 48 tests · live registry, router, trace



answer card and live trace — untrained tools carry an honest HEURISTIC badge instead of a fake number

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea · potential challenges and risks · strategies for overcoming these challenges

15-DAY PATH

- **5 Sep** Round-1 pitch
- **D1–D4** data + M1 QLoRA starts
- **D5–D8** M4, M3, M7 into registry
- **D9–D12** planner, abstention, report
- **D13–D15** benchmarks + demo video
- **20 Sep** code + models + test

FEASIBILITY

- **Weights are open, cost is zero:** Qwen2-VL, Grounding DINO-T, BiT, UNet++, BigEarthNet v2.0 checkpoints — no licence, no paid API.
- **Scope was cut to fit the compute:** 4-bit QLoRA on a 2B backbone fits a free 16 GB notebook GPU; M4 and M3 train on a 6 GB laptop.
- **The skeleton is already code, not a plan:** ingestion, registry, guardrail, router, trace and web UI run today with 48 passing tests; new tools swap in behind a frozen contract.

VIABILITY & PRACTICALITY

- **Deployable where ISRO actually is:** dockerised, single node, no internet, CPU fallback for the light path; ingest accepts Cartosat-2S and RISAT pairs by design (P14).
- **Every gate is measurable:** metric thresholds on public splits before submission, and scores normalised before combining (P15) so one spiked metric cannot hide a weak one.
- **Demo-safe by construction:** deterministic router plus cached-answer mode means the presentation never depends on a lucky inference.

WHAT IS ALREADY PROVEN

- **Optical / SAR / fused numbers exist today:** BE v2.0 resnet50 checkpoints — macro mAP 0.711 on S1+S2 against single-modality variants, at zero training cost.
- **Data is verified, not assumed:** BigEarthNet.txt = 467 MB parquet, 9.55 M rows, CDLA-Permissive-1.0; BE v2.0 imagery 118 GB plus 282 MB per-pixel reference maps.
- **Risk work is done:** all 15 binding PS clauses mapped to a design decision; 5 gaps in the official text logged (G1–G5).

	POTENTIAL CHALLENGES AND RISKS	STRATEGIES FOR OVERCOMING THESE CHALLENGES
01	Domain gap — we train on Sentinel (10–20 m), the hidden ISRO set is Cartosat-2S (~1 m) and RISAT L-band (G5).	Tile-based, sensor-agnostic pipeline; week-1 proxy runs on public VHR and L-band sets; report per-class scores, not only the mean.
02	6 GB VRAM at the venue; no GPU is promised to teams.	2B backbone at 4-bit, load-and-evict per tool, no vLLM; heavy training only on free 16 GB notebooks.
03	The PS judging table is an unfilled placeholder (G1) and reference annotations stay undisclosed (P14).	Train on the five PS-named public benchmarks, freeze contracts early, and ship a normalised metric table instead of one lucky number.
04	Demo-day failure in front of the panel.	Deterministic fallback router, cached-answer kill switch, 3–5 frozen demo scenes, dry run on the presenting laptop.
05	Three trained models in 15 days is tight.	Binding triage M1 / M4 / M3 with free pretrained M7, checkpoint every 15 min, one clear day of slack before submission.

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IMPACT AND BENEFITS

Potential impact on the target audience · benefits of the solution (social, economic, environmental)

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POTENTIAL IMPACT ON THE TARGET AUDIENCE · who can ask in minutes what today needs a GIS team

SatQuery AI
one question
one trace
six user groups

ISRO / SAC analysts
a tasking question answered in minutes with the evidence attached, instead of days of scripting in QGIS

Disaster & flood cells
change map plus change-VQA on the first pass after an event; SAR works through cloud and at night

Agriculture & water boards
reservoir, crop and water-body questions in plain language, so field officers stop waiting

Urban planning & municipal
built-up growth measured from an auditable mask, defensible in an encroachment case

Researchers & students
open traces, published metric tables and a downloadable report make results reproducible

National capability
an in-house replacement for single-task vendor imagery pipelines: Rs 0 licence, offline, on Indian data

BENEFITS OF THE SOLUTION

SOCIAL
Democratises satellite intelligence: an officer types a question and gets a checkable answer. The skill barrier, not the satellite, is what is removed.

plain language · answers legible to non-experts · no vendor lock-in

ECONOMIC
One assistant replaces a shelf of single-task imagery pipelines, and returns analyst-hours on every tasking cycle.

Rs 0 licence · days to minutes per query · trains on a 6 GB GPU

ENVIRONMENTAL
Faster detection of floods, shrinking water bodies and encroachment — the response starts while the window is still open.

cloud-proof SAR · change flagged within one revisit

SCALES WITHOUT A REWRITE
One tool contract per capability, so a new sensor or band combination is a registry entry rather than a rebuild — and every answer stays auditable.

"Every answer with evidence, confidence and an audit trail: satellite intelligence anyone can query, offline, at zero licence cost." Pilot path: SAC internal pilot » ISRO operational nodes » state disaster cells and imagery desks.

RESEARCH AND REFERENCES

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Details / links of the reference and research work · datasets, models, benchmarks and the evidence we gathered ourselves

OUR PRIMARY RESEARCH FOR THIS SUBMISSION

- **Clause-by-clause PS reading:** all 15 binding clauses (P1–P15) mapped to a design decision, and 5 gaps in the official text logged rather than guessed — including the unpublished judging table and the truncated dataset links.
- **Data verified at source:** BigEarthNet.txt = 467 MB parquet, 9.55 M rows, CDLA-Permissive-1.0, joined to BE v2.0 imagery (118 GB) and its 282 MB per-pixel reference maps by patch_id.
- **Zero-shot vs adapted table:** the compliance proof for P12, filled with reproduced training logs before 20 Sep — not with claims.
- **Working thin prototype:** GDAL ingestion, registry.yaml + GET /tools, guardrail, router, SQLite trace, web UI, report export — 48 pytest tests, CPU-only, offline.

DATASETS AND BENCHMARKS NAMED BY THE PS

BigEarthNet.txt	primary adaptation data for image-text representations over multisensor imagery · arXiv:2603.29630 · huggingface.co/datasets/BIFOLD-BigEarthNetv2-0
VRSBench	single-image captioning, grounding and VQA · NeurIPS 2024 Datasets & Benchmarks · arXiv:2406.12384 · github.com/lx709/VRSBench
RSVQA	large-scale single-image VQA · Lobry et al., IEEE TGRS 2020
CDVQA	bi-temporal change VQA and semantic change maps · arXiv:2112.06343 · github.com/YZHJessica/CDVQA
TAMMI · LEVIR-CD · WHU-CD	VHR RGB + S1 + S2 triplets for cross-modal VQA; change-mask volume · github.com/justchenhao/Levir-CD

EVALUATION PROTOCOL · gates to clear before 20 Sep

MANDATE	BENCHMARK	METRICS · gates, normalised per P15
Caption / scene	VRSBench test	CIDEr · BLEU-4 · METEOR · ROUGE-L
Text-guided grounding	VRSBench refs	box IoU ≥ 0.5 · COCO AP
Single-image VQA	RSVQA + VRSBench	exact match · soft accuracy
Change-VQA	CDVQA test	exact match per answer category
Change map	CDVQA · LEVIR-CD	IoU · F1 · mIoU, per class

Latency on a 6 GB laptop: single-image 10 s · grounding 8 s · change 20 s · cross-modal 20 s, recorded per step.

MODELS, METHODS AND OFFICIAL SOURCES

- **Qwen2-VL-2B** (arXiv:2409.12191) — native multi-image VLM backbone; adapted with QLoRA (arXiv:2305.14314) on 4-bit.
- **Grounding DINO** (arXiv:2303.05499) — text-to-box base for the mandatory grounding task (P3).
- **BiT and UNet++** (arXiv:1807.10165) — change and land-cover heads, initialised from BigEarthNet v2.0 S1 / S2 / S1+S2 checkpoints.
- **GDAL / rasterio · COG tiling** — geospatial correctness: CRS, band order, co-registration offset, 512 px windows.
- **Official sources** — sih.gov.in (PS 26167, ISRO, and the idea-PPT format) · isro.gov.in and the SAC evaluation-set note (P14) · BigEarthNet v2.0, Zenodo record 10891137.

MANDATE COVERAGE · every mandatory clause in PS 26167 and where it is handled

- **Single-image VQA** ✓ — mandatory baseline · M1 on RSVQA/VRSBench
- **Caption + grounding** ✓ — both shipped · M1 caption + M3 boxes (P3)
- **Change + change-VQA** ✓ — M4 map + M1 two-image on CDVQA (P4)
- **Optical + SAR jointly** ✓ — 2-image mode + S1/S2/fused ablation (P5)
- **Registry + parameters** ✓ — registry.yaml + guardrail whitelist (P7–P8)
- **Auditable trace** ✓ — trace.json + PDF report, per step (P9)